OLYMPIAD · EVERYDAY MATHEMATICS · OLYMPIAD PREPARATION · SHORT NOTES

Chapter 12: Simple and Compound Interest

Topic-wise short notes with solved examples

Interest on borrowed money, simple interest by the formula, compound interest by repeated growth, and the quick shortcut for the difference in two years.

1 Simple interest

- **Interest** is the extra money paid for borrowing. The money borrowed is the **principal** P, the yearly **rate** is $R\%$ and the **time** is T years.
- In **simple interest** the interest is always worked out on the original principal.

$$\text{SI} = \frac{P \times R \times T}{100} \qquad A = P + \text{SI}$$

- A is the **amount**: what has to be paid back in all.
- If any three of P, R, T and SI are known, the fourth can be found by rearranging the formula.

Example 1.1 Find the simple interest on Rs 5000 at 8% per year for 3 years.

SOLUTION

$\text{SI} = \frac{5000 \times 8 \times 3}{100}$.

Rs 1200

Example 1.2 A sum of Rs 2000 earns a simple interest of Rs 600 in 3 years. Find the rate of interest.

SOLUTION

$600 = \frac{2000 \times R \times 3}{100}$, so $R = \frac{600 \times 100}{6000}$.

10% per year

2 Compound interest

- In **compound interest** the interest of each year is added to the principal, and the next year's interest is worked out on the larger amount.
- After n years: $A = P\left(1 + \frac{R}{100}\right)^n$ and $\text{CI} = A - P$.
- If the interest is added **half-yearly**, use half the rate and double the number of periods: $A = P\left(1 + \frac{R/2}{100}\right)^{2n}$.

- For depreciation (the value falls each year) use $\left(1-\frac{R}{100}\right)^{n}$.

Example 2.1 Find the compound interest on Rs 10 000 at 10% per year for 2 years.

SOLUTION

$A = 10\,000 \times 1.1 \times 1.1 = 12\,100.$

$\mathrm{CI} = 12\,100 - 10\,000.$

Rs 2100

Example 2.2 Find the amount when Rs 8000 is invested for 1 year at 10% per year, interest added half-yearly.

SOLUTION

Rate per half-year $= 5\%$ and the number of half-years $= 2$.

$A = 8000 \times 1.05 \times 1.05.$

Rs 8820

Example 2.3 A machine worth Rs 50 000 loses 10% of its value every year. Find its value after 2 years.

SOLUTION

$50\,000 \times 0.9 \times 0.9.$

Rs 40 500

3 Comparing the two

- In the **first** year simple interest and compound interest are equal. After that, compound interest is larger.
- For **2 years**, $\mathrm{CI} - \mathrm{SI} = P\left(\frac{R}{100}\right)^{2}$.

Example 3.1 Find the difference between the compound interest and the simple interest on Rs 5000 at 10% per year for 2 years.

SOLUTION

$5000 \times \left(\frac{10}{100}\right)^{2} = 5000 \times 0.01.$

Rs 50

REMEMBER

For compound interest over 2 or 3 years, multiply year by year. You can see each step, and mistakes with powers are avoided.

MathSetu